Course Description

Critical Thinking & Problem Solving in the Age of AI:

"That is not only not right; it is not even wrong!" – Wolfgang Pauli

Artificial intelligence can't replace thinking, but it can amplify poor reasoning. As Ai systems become increasingly capable of generating plausible explanations, analyses, and recommendations, the ability to evaluate evidence, identify hidden assumptions, and recognize uncertainty becomes more important.

This four-day workshop develops the intellectual tools and habits of mind required for rigorous, evidence-based problem solving. Participants explore fundamental questions about knowledge, reasoning, scientific inquiry, and decision-making, with particular attention to the ways cognitive biases, flawed assumptions, misleading data, and Ai-generated information can undermine otherwise sophisticated thinking.

The course integrates four complementary frameworks:

- **Kahneman's System 1/System 2 model:** Recognizing when intuition and cognitive shortcuts may lead us astray.
- **The Paul–Elder Critical Thinking Framework:** Examining the structure, assumptions, and quality of our reasoning.
- **Polya's Four-Step Method:** Transforming complex and ambiguous problems into manageable, systematic investigations.
- **Wild and Pfannkuch's PPDAC Cycle:** Understanding how data, research design, analysis, and interpretation shape scientific conclusions.

Participants will not just learn the philosophy of how to think; they will diagnose why they fail. Throughout the course, participants return to one central question:

How do we know when to trust our own conclusions?

Course Learning Outcomes

1. Evaluate knowledge and Uncertainty: Distinguish among types of uncertainty, explain how new evidence should influence confidence, and recognize when conclusions exceed their justification.
2. Diagnose Cognitive Failure: Identify and articulate the root causes of error in scientific and clinical judgment, distinguishing between System 1 heuristics and System 2 analytics, especially wrt cognitive biases, logical fallacies, hidden assumptions.
3. Apply Frameworks of Thought: Apply the Paul-Elder Framework to decompose complex, ambiguous problems into their constituent elements and evaluate them against universal intellectual standards.
4. Evaluate and Apply Statistical Intuition: Identify measurement error, selection bias, confounding, and aggregation problems; distinguish association, prediction, and causation; and evaluate the limitations of scientific methods.
5. Evaluate Ai-generated reasoning and recommendations: Identify unsupported claims, methodological weaknesses, uncertainty, and potential consequences in AI-assisted analyses.
6. Develop strategies for improved judgment: Apply cognitive forcing tools, pre-mortems, and reflective practices to create a personal protocol for making better decisions under uncertainty.

Schedule & Content

Day	Topic & Content	Learning Outcomes
	MODULE 1: Knowledge, Uncertainty, and Calibration ***Q:*** *Why do smart people make predictable mistakes?*	

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A. Explanation of the Course (10 min): **The Calibration problem** (when to trust our own conclusions?)
B. Thinker's log (5 min)
   - **Prompt:** What did I believe confidently this week that turned out to be wrong or uncertain?
C. Activity 1 (20 min): **The Crisis of Authority Discussion**
   - When have experts been confidently wrong?
   - What makes a source of knowledge reliable?
D. Lecture 1 (25 min): **What is knowledge and Why is it 'breaking"?**
   - Justified true belief and the Gettier problem
   - Epistemic humility
   - Tacit knowledge and the limits of expertise

Break (10 min)

E. Activity 2 (25 min) : Bayesian updating and the medical diagnosis problem.
   - Explore base-rate neglect, prior beliefs, new evidence, and the relationship between evidence and confidence
F. Lecture 2 (25 min): **System 1/System 2 – Kahneman**
   - Cognitive Reflection Test (CRT)
   - Intuitive versus deliberative decision
   - Exposing hidden assumptions and proxy variables
G. Activity 3 (25 min): **Hiring Bias Game**

Break (10 min)

H. Lecture 3 (30 min): **Types of uncertainty and cognitive "forcing":**
   - Aleatoric, epistemic, and model uncertainty
   - Overconfidence, motivated reasoning, and strategies to interrupt automatic thinking
   - Introduction to the 4 frameworks
I. Case Study (25 min): Hospital Ai adoption- Part 1.
   - Make an initial recommendation about adopting an Ai diagnostic system, record confidence in that recommendation, and identify information needed to justify it. What evidence would change your recommendation?

Homework:

- Facione's "Critical Thinking: What it is and Why it Counts".
- Be prepared to discuss difference between a critical thinker and a non-critical thinker
- Begin Person Heuristic Protocol: identify one cognitive bias or reasoning habit that you will address

Summary of the module:

This module introduces the epistemological foundations of critical thinking. Participants explore what constitutes knowledge, why expertise does not eliminate error, and how cognitive biases and uncertainty affect judgment. Particular attention is given to the distinction between confidence and accuracy, and to the importance of updating beliefs when confronted with new evidence.

By the end of this module, you will be able to:

1. Articulate the epistemological question at the heart of the course: **how do we know what we know, and how do we catch ourselves when we're wrong?**
2. What constitutes knowledge? Distinguish between belief, justification, and knowledge and explain why a correct conclusion does not necessarily imply sound reasoning.
3. Distinguish between System 1 (fast/heuristic) and System 2 (slow/algorithmic) thinking and identify examples of each failing in professional contexts.
4. Recognize cognitive bias as a system design failure, not a personal character flaw, and identify one structural intervention that could mitigate it.
5. Preview the four frameworks of the course and make an initial hypothesis about which would address a specific reasoning failure.

MODULE 2: The Architecture of Reasoning

Q: *How do we deconstruct complex problems to make them tractable and expose hidden assumptions?*

2	A. Thinker's log (5 min) • **Prompt:** Think of a belief you hold strongly in your work. What evidence would it take to change your mind? B. Homework debrief (10 min): Share-Pair C. Lecture 1 (20 min): **Structure of an argument & fallacies** • Premises, warrants, conclusions, deductive, inductive, and abductive reasoning • Validity and soundness D. Activity 1 (25 min): **Wason Selection Task** • Confirmation bias, falsification, and the difficulty of testing conditional claims E. Lecture 2 (20 min): **Paul-Elder Framework** • 8 elements of thought • Universal intellectual standards for evaluating reasoning **Break (10 min)** F. Activity 2 (30 min): **Intellectual Autopsy** • Evaluate a retracted paper using Paul-Elder • Identify weaknesses in assumptions, evidence, inference, and institutional processes G. Activity 3 (20 min): **Unpack your brain** • Apply Paul-Elder to a problem or conclusion from participants' own work • Compare stated assumptions with actual operating assumptions **Break (10 min)** H. Lecture 3 (30 min): **Polya's Four Step Method and Fermi Estimation** • Practice defining a problem, and developing a strategy • Estimating with incomplete information, and evaluating the result I. Case Study (25 min): Hospital Ai adoption- Part 2. Introduce Popperian falsification and the distinction between testing and confirming hypotheses. Audit your Hospital Ai adoption initial recommendation with Paul-Elder and Polya. **Homework:** • Identify the weakest element in your reasoning and consider how it could be improved. • Update the Personal Heuristic Protocol with reasoning and problem-solving safeguards. • Re-read one of your own recent abstracts, grant aims or other and map it to the 8 elements. Identify the weakest element. Apply the eight elements of Paul–Elder to an argument, abstract, or decision from your own work.	**Summary of the module:** This module examines how arguments are constructed and evaluated, and how structured approaches can improve problem solving. Participants learn to distinguish types of scientific inference, identify logical and methodological weaknesses, and apply the Paul–Elder and Polya frameworks to real scientific problems. **By the end of this module, you will be able to:** 1. Distinguish deductive, inductive, and abductive reasoning, and recognize their complementary roles in scientific inquiry. 2. Identify premises, warrants, conclusions, and weaknesses in scientific arguments. 3. Explain falsification and distinguish evidence that supports a claim from evidence that genuinely tests it. 4. Apply the Paul–Elder Framework to evaluate a scientific argument and expose unexamined assumptions. 5. Use Polya's method to decompose an ambiguous problem into tractable components and develop a defensible plan. 6. Recognize how scientific incentives, group reasoning, and institutional assumptions can contribute to persistent errors.
	MODULE 3: Data & Thinking Probabilistically ***Q**: How do data, models and methodological choices deceive us, and how can we uncover the truth?*	
3	**A.** Thinker's log (5 min) • **Prompt:** Describe a time when a number or statistic changed your mind. Was the change justified? **B.** Homework debrief (5 min): Share-Pair **C.** Activity 1 (20): **The Assumption Auction.** • Economists gauge people's confidence in belief by betting on them. We'll do the same with plausible claims.	**Summary of the module:** This module extends critical reasoning to scientific data. Participants examine the process through which observations become data, how experimental and statistical decisions affect inference, and why data

	• This is intended to reveal hidden assumptions. **D.** Lecture 1 (20 min): **Epistemology of data** • Data are representations of reality, not reality • Biography of a number, measurement error, sampling bias, and aggregation **E.** Lecture 2 (25 min): **Scientific inference and causation.** • Association versus prediction versus causation • Confounding, experimental control, randomization • Limits of observational evidence **Break (15 min)** **F.** Lecture 3 (25 min): **Wild & Pfannkuch PPDAC cycle** • Domain knowledge and iterative reasoning **G.** Activity 2 (25 min): **Simpson's Paradox** • Analyze aggregated and stratified treatment outcomes • Identify confounding and interpretive limitations **Break (15 min)** **H.** Activity 3 (25 min): **Critical Control Points** • Trave a biomedical dataset (or scientific claim) through its lifecycle, identifying where methodological choices introduce bias and/or uncertainty. **I.** Case Study (30 min): **Hospital Ai adoption**- Part 3. Apply PPDAC to evaluate the system's training data, target population, performance metrics, confounding, external validity, and proposed clinical use. Revise the recommendation and confidence assessment. **Homework:** • What assumption did you make this week, in your research or clinical work, about the data you were working with? Was it justified? • Add data interrogation questions to your Personal Heuristic Protocol. • Bring one real dataset, figure, or table from your own work (or a published paper) to Day 4. Come with a question about it	analysis alone cannot compensate for flawed research design. The module emphasizes the distinction between association, prediction, and causation, together with the iterative nature of scientific inquiry. **By the end of this module, you will be able to:** **1.** Explain why data are imperfect representations of reality and identify the decisions underlying their production. **2.** Distinguish measurement error, selection bias, confounding, and aggregation effects. **3.** Distinguish association, prediction, and causation and explain how research design affects the strength of causal claims. **4.** Apply the PPDAC cycle to evaluate scientific evidence and recognize the iterative nature of statistical inquiry. **5.** Explain Simpson's Paradox and identify situations where aggregated results conceal meaningful differences among subgroups. **6.** Identify critical control points in a research or analytical workflow where additional scrutiny or methodological changes are warranted.
	MODULE 4: The Examined Machine: Problem solving in the age of Ai Q: How do we ensure that our decisions, and those supported by Ai, are justified by the evidence?	
4	A. Thinker's log (5 min) • **Prompt:** What patterns have emerged in your thinking across the first three days? B. Homework debrief (5 min): Share-Pair C. Lecture 1 (20 min): **Epistemology of Ai-Assisted Reasoning.** • Plausibility versus validity • Automation bias • The illusion of understanding D. Activity 1 (40 min): **The Turing Audit** • Evaluate an Ai-generated scientific literature summary using the four frameworks. • Identify unsupported conclusions, hidden assumptions, statistical weaknesses, and overconfidence E. Lecture 2 (10 min): **Comparison of four frameworks** • What type of failure did each framework excel in exposing?	**Summary of the module:** The final module integrates the four frameworks and applies them to complex, AI-assisted decisions. Participants examine the distinction between plausible explanations and trustworthy conclusions, evaluate the limits of AI-assisted reasoning, and consider the ethical and practical consequences of delegating analytical work to machines. **By the end of this module, you will be able to:** **1.** Explain why plausible, coherent AI-generated explanations may not constitute reliable scientific reasoning.

Break (10 min)

F. Activity 2 (40 min): **The Pre-mortem**
- Evaluate a proposed Ai-assisted sepsis detection system (PredictAI).
- Imagine that the project has failed and identify cognitive, reasoning, methodological, and institutional causes.

G. Lecture 3 (15 min): **Responsible Ai and Delegation**
- Ai delegation mapping (Anthropic framework)
- Human oversight
- Ethical consequences of model and data choices

Break (10 min)

H. Activity 3 (30 min): **Personal Heuristic Procol**
- Develop and Pair-Share a practical checklist or decision process incorporating cognitive safeguards, assumption testing, evidence evaluation, and falsification checks.

I. **Case Study:** Hospital Ai adoption (Final decision):
- Revisit initial recommendation
- Identify how and why it changed
- Determine what further evidence is necessary

J. **Course synthesis** (10 min): Return to the calibration problem and identify one change to implement in professional practice.

Homework:
- A peer-reviewed Personal Heuristic Protocol
- Revised, final hospital Ai adoption recommendation

2. Apply the four frameworks to identify weaknesses in AI-generated analyses and recommendations.
3. Evaluate when AI delegation is appropriate based on task characteristics, domain expertise, uncertainty, and potential consequences.
4. Conduct a pre-mortem to anticipate reasoning, data, and implementation failures in a proposed project.
5. Recognize how ethical considerations, institutional incentives, and data limitations influence the reliability of AI-assisted decisions.
6. Develop a personal, evidence-informed protocol for evaluating claims, managing uncertainty, and reducing predictable reasoning failures.

Resources

Resource	Author	Notes
Thinking, Fast and Slow	*Daniel Kahneman*	Foundational text for System 1/2. Wikipedia summary available free.
The Checklist Manifesto	*Atul Gawande*	Core reading for cognitive forcing strategies and structural interventions.
Calling Bullshit	*Bergstrom & West*	Free course at callingbullshit.org. Book also available. Information literacy for the data age.
Critical Thinking: What It Is and Why It Counts	*Peter Facione*	Free PDF. Assigned reading for Day 1 homework.
Weapons of Math Destruction	*Cathy O'Neil*	Algorithmic bias and real-world consequences of statistical misuse.
What Is Statistical Thinking? (1998)	*Wild & Pfannkuch*	Primary source for the PPDAC cycle. Free PDF available.
Death of a Knowledge System	*Nicholas Michaelsen*	Substack essay. Contextualizes the Day 1 crisis of authority discussion.
Anthropic Economic Index (2026)	*Anthropic*	Identifies skills most augmented vs. displaced by AI.
On the Measure of Intelligence (2019)	*François Chollet*	Foundational for Day 5 synthesis on intelligence and generalization. Free on arXiv.
How to Solve It	*George Polya*	Primary source. Free PDF available via math.hawaii.edu.
Paul-Elder Critical Thinking Mini-Guide	*Paul & Elder*	Free PDF. Reference for all Day 2 content.

Resource	Author	Notes
Procedures to Change Implicit Measures (2019)	*Forscher et al.*	Meta-analysis of bias intervention durability. PubMed free access.
Illusions of Understanding (2024)	*Crockett & Messeri*	Nature paper on AI and epistemic over-reliance. Relevant to Day 5.
Habits of Mind	*Systems Thinker*	Free article on disciplined choice-making heuristics.
Data Science Ethics	*Carlo Vega*	Despite its name, this is mostly focused on "What is knowledge?" and "What is sufficient evidence?". https://hdsr.mitpress.mit.edu/pub/5to0suwj/release/2 He has a course here with activities: https://posde.cvega.es/
Recipes for Science	*Potochnik, Colombo, Wright*	Each chapter corresponds to a new philosophical framework for science: including, deductive, inductive, abductive, statistical, Bayesian, causal, and society.
The Pyramid of AI Cognitive Roles	*Inara Scott*	LinkedIn article. I hate that I am recommending a LinkedIn article, but.... Useful framing for AI collaboration hierarchy.